

Quantification of generalised measurement incompatibility of mutually unbiased bases

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Outline

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- 3 The joint-measurability SDP
- 4 Compatibility hypergraphs
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Why measurement incompatibility?

Classical physics. Every set of observables can in principle be measured jointly.

Quantum mechanics. Some sets of measurements are **incompatible**: no single experiment can reproduce the outcome statistics of all of them.

- First recognised through non-commutativity $AB \neq BA$ and uncertainty relations.
- Modern, information-theoretic viewpoint: incompatibility is a **resource**.

Incompatible measurements are necessary for

- Bell nonlocality, EPR steering, quantum contextuality
- Quantum key distribution, state discrimination, random access codes

Goal of this project

Quantify the noise robustness of incompatibility for **mutually unbiased bases (MUBs)**, and extend the quantification to richer compatibility structures.

Mutually unbiased bases (MUBs)

Two orthonormal bases $\mathcal{B} = \{|b_i\rangle\}$, $\mathcal{B}' = \{|b'_j\rangle\}$ of $\mathbb{C}^d$ are **mutually unbiased** iff

$$|\langle b_i | b'_j \rangle|^2 = \frac{1}{d} \quad \forall i, j.$$

Preparing in any state of one basis $\Rightarrow$ uniformly random outcome in the other: the two measurements are **maximally non-commuting**.

- $d = 2$: the eigenbases of $\sigma_X, \sigma_Y, \sigma_Z$ form 3 MUBs.
- In general at most $d + 1$ MUBs exist, achieved when d is a prime power (*Wootters & Fields, 1989*).
- For $d = 6$ no fourth MUB has ever been found — a long-standing open problem.

MUBs are the prototypical maximally incompatible measurements.

Joint measurability and white-noise robustness

Joint measurability. A family $\{M_{a|x}\}$ of POVMs is **jointly measurable** iff there exists a single parent POVM $\{G_\lambda\}$ and conditional probabilities $p(a | x, \lambda)$ such that

$$M_{a|x} = \sum_{\lambda} p(a | x, \lambda) G_{\lambda} \quad \forall x, a.$$

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White-noise robustness. Mix each measurement with noise at level $\eta \in [0, 1]$:

$$M_{a|x}^{\eta} = \eta M_{a|x} + (1 - \eta) \operatorname{Tr}(M_{a|x}) \frac{I}{d}.$$

The figure of merit is the critical noise parameter

$$\eta^{\star} = \max\{\eta : \{M_{a|x}^{\eta}\} \text{ is jointly measurable}\}.$$

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$$\eta^* = \max\{\eta : \{M_{a|x}^{\eta}\} \text{ is jointly measurable}\}.$$

Key fact

η^* can be computed exactly by a **semidefinite program (SDP)**.

The standard SDP, illustrated on the qubit

Two noisy Pauli measurements on $\mathbb{C}^2$, $M_{a|X}^\eta$ and $M_{a|Z}^\eta$, $a \in \{+, -\}$.

$$\begin{aligned} & \text{maximise} \quad \eta \in [0, 1] \\ & \text{s.t.} \quad G_{\lambda_X \lambda_Z} \succeq 0, \quad \sum_{\lambda_X, \lambda_Z} G_{\lambda_X \lambda_Z} = I, \\ & \quad \sum_{\lambda_Z} G_{a \lambda_Z} = M_{a|X}^\eta, \quad \sum_{\lambda_X} G_{\lambda_X a} = M_{a|Z}^\eta. \end{aligned}$$

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Optimal value

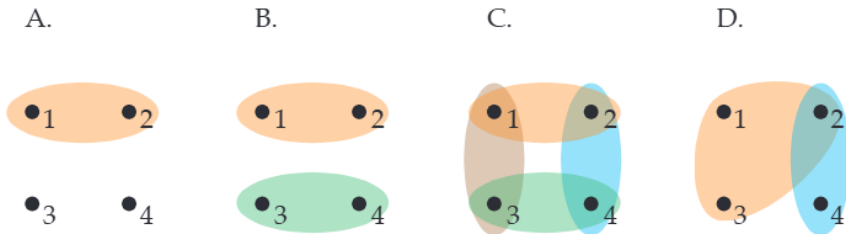
$$\eta^*(2, 2) = \frac{1}{\sqrt{2}} \approx 0.7071.$$

Matches the $k = 2, d = 2$ entry of Table I of Designolle *et al.*, PRL 122, 050402 (2019).

Compatibility hypergraphs (Quintino *et al.*)

For ≥ 3 measurements, compatibility is no longer yes/no but **structured**: triples can be pairwise jointly measurable yet fail to be jointly measurable as a whole.

Compatibility hypergraph $\mathcal{C} = [C_1, \dots, C_N]$: each hyperedge lists a subset of measurements required to be jointly measurable.



Examples on 4 measurements; reproduced from Quintino *et al.*, arXiv:1902.05841.

Our generalisation: hypergraph-parameterised SDP

We implement the SDP for an **arbitrary** compatibility hypergraph $\mathcal{C}$:

- In each branch i , the measurements indexed by C_i must be jointly measurable;
- Measurements outside C_i are arbitrary POVMs in that branch.

Implemented as the `robustness_hypergraph` routine in CVXPY.

`hypergraph_compatible(k)` $\mathcal{C} = \{(0, 1, \dots, k-1)\}$
 $\Rightarrow$ **full k -wise joint measurability**; recovers Designolle's threshold.

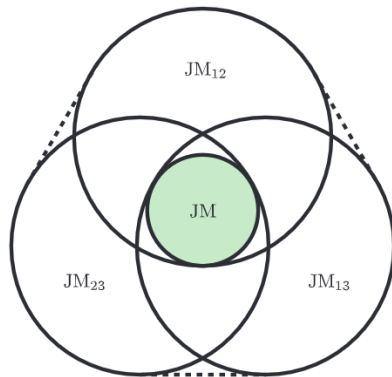
`hypergraph_pair(k)` $\mathcal{C} = \{(i, j) : 0 \leq i < j \leq k-1\}$
 $\Rightarrow$ **all-pairs** $((k-1)$ -wise) compatibility, the natural pairwise relaxation.

Geometric picture: convex hull of pairwise families

A family is **pairwise-compatible** iff it lies in the convex hull of the single-pair jointly measurable families:

$$M_{a|x}^{\eta} = \sum_{i < j} p_{ij} M_a^{(x),(i,j)}.$$

- Each branch jointly measures *one* pair; the remaining $k - 2$ measurements are arbitrary.
- Strictly weaker than full JM: $\eta_{\text{pairs}}^* \geq \eta_{\text{JM}}^*$.
- $\text{JM} \subset \text{conv}(\text{JM}_{12}, \text{JM}_{13}, \text{JM}_{23})$.



Adapted from Quintino *et al.*

JM : fully jointly measurable families.

JM_{ij} : pairs (i, j) jointly measurable, the third arbitrary.

Analytical lower bound (our contribution)

Conjecture: the optimal pairwise mixture is the **uniform** one, $p_{ij} = 1/\binom{k}{2}$.

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Branch counting. By MUB symmetry, every setting x appears in exactly $k - 1$ *active* branches (at noise level $\eta^*(2, d)$) and in $\binom{k}{2} - (k - 1)$ *inactive* branches (at noise level 1).

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Affineness lemma.

$$\eta_{(k-1)}^* = \frac{1}{\binom{k}{2}} \left[(k-1) \eta^*(2, d) + \frac{(k-1)(k-2)}{2} \right].$$

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Substitute Designolle's $\eta^*(2, d) = (2 + \sqrt{d})/[2(\sqrt{d} + 1)]$ and simplify:

Closed-form lower bound

$$\eta_{(k-1)}^*(k, d) = 1 - \frac{\sqrt{d}}{k(\sqrt{d} + 1)}$$

Table I: reproducing full joint measurability

k	d			
	2	3	5	7
2	0.7071	0.6830	0.6545	0.6371
3	0.5774	0.5686	0.5393	0.5101
4	—	0.5000	0.4615	0.4436
5	—	—	X	X
6	—	—	0.4082	X
8	—	—	—	0.3536

Every closed-form entry of Designolle *et al.* (2019) reproduced to within 10^{-4} .
“X” = omitted, $d^k > 3000$ (intractable on a laptop CPU).

Table II: new results, pairwise ($(k-1)$ -wise) compatibility

k	d				
	2	3	5	7	11
2	0.7071	0.6830	0.6545	0.6371	0.6158
3	0.8047	0.7887	0.7697	0.7581	0.7439
4	—	0.8415	0.8273	0.8186	0.8079
5	—	—	0.8618	0.8549	0.8463
6	—	—	0.8848	0.8791	0.8719
7	—	—	—	0.8963	0.8902
8	—	—	—	0.9093	0.9040
9	—	—	—	—	0.9146

Every entry matches the analytical bound to solver precision $\Rightarrow$ **tightness**.
 SDP scales as $\mathcal{O}(k^3 d^2)$, fast even for $d = 11$.

The qualitative punchline: a monotonicity flip

Observation

- Full joint measurability: $\eta_{\text{JM}}^*(k, d)$ is **decreasing** in k , $\rightarrow 0$ as $k, d \rightarrow \infty$.
- Pairwise compatibility: $\eta_{(k-1)}^*(k, d) = 1 - \frac{\sqrt{d}}{k(\sqrt{d} + 1)}$ is **increasing** in k , $\rightarrow 1$ as $k \rightarrow \infty$.

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Intuition. In the pairwise SDP, any fixed measurement appears in only $k - 1$ of the $\binom{k}{2}$ branches — a vanishing fraction $2/k$. Its “responsibility” dilutes as k grows, so the relaxation becomes nearly vacuous.

Quantitative incarnation of the Quintino hierarchy: finer hypergraphs $\Rightarrow$ looser constraint $\Rightarrow$ larger η^* .

Summary of contributions

- 1 Reproduced the standard joint-measurability noise-robustness SDP for MUBs in prime dimensions; matches Designolle *et al.* to 10^{-4} .
- 2 Generalised the program to **arbitrary compatibility hypergraphs**, with two canonical helpers `hypergraph_compatible` and `hypergraph_pair`.
- 3 Derived a **closed-form lower bound** $\eta^* = 1 - \sqrt{d}/[k(\sqrt{d} + 1)]$ for the pairwise (($k-1$)-wise) relaxation.
- 4 Verified numerically that the bound is **tight** for every (k, d) computed.
- 5 Exposed the **monotonicity flip** between full JM and pairwise relaxation, quantifying the Quintino hierarchy.

Going further

- (i) **Tightness certificate.** Implement the **dual SDP**: either prove tightness analytically by exhibiting a feasible dual witness, or identify the regime where the equal-weight pairwise mixture is sub-optimal.
- (ii) **Intermediate hypergraphs.** Between full JM and all-pairs lies a rich lattice — all triples, $(k-2)$ -subsets, asymmetric scenarios. Our framework handles them with no modification.
- (iii) **Composite dimensions.** For $d = 6$, no fourth MUB has ever been found. Running the same hypergraph SDP on the three known MUBs in $d = 6$ may provide indirect evidence on this longstanding open question.

Thank you!

Questions?